Question Bank
Module-5

Q.1. Three fuel injection systems are tested for efficiency and the following coded data are obtained.

System 1	System 2	System 3
48	60	57
56	56	55
46	53	52
45	60	50
50	51	51

Do the data support the hypothesis that the systems offer equivalent levels of efficiency?

Q.2. A vehicle manufacturer wishes to test the ability of three types of steel-alloy panels to resist corrosion when three different paint types are applied. Three panels with differing steel-alloy composition are coated with three types of paint. The following coded data represent the ability of the painted panels to resist weathering.

Paint Type	Steel-Alloy 1	Steel-Alloy 2	Steel-Alloy 3
1	40	51	56
2	54	55	50
3	47	56	50

Use a two-way ANOVA procedure to determine whether any difference in the ability of the panels to resist corrosion may be assigned to either the type of paint or the steel-alloy composition of the panels.

Q.3. A nutritionist wants to compare the effectiveness of three different diet plans on weight loss. She selects three groups of participants, each following one of the diets for a month. At the end of the month, their weight loss (in kg) is recorded as follows:

- Diet A: 3.2, 4.1, 3.8, 4.5, 3.9
- Diet B: 5.1, 4.9, 5.3, 5.0, 5.2
- Diet C: 2.8, 3.0, 2.5, 2.9, 2.7

Using One-Way ANOVA, determine if there is a significant difference in weight loss among the three diet plans.

Q.4. A device has a constant failure rate of **0.02** failures per hour. Find:

(a) The reliability function R(t)

(b) The hazard rate function h(t)

Q.5. The failure density function of the random variable T is given by

$$f(t) = \begin{cases} 0.012e^{-0.012t}, & t \geq 0 \\ 0, & elsewhere \end{cases}$$

Calculate

(i) Reliability of the component.

(ii) Reliability of the component for a 100 hour mission time.

(iii) Mean time to failure (MTTF)

(iv) What is the life of the component if a reliability of 0.96 is desired?

Q6.

Illustration 2: To test the significance of variation in the retail prices of a commodity in three principal cities, Mumbai, Kolkata, and Delhi, four shops were chosen at random in each city and the prices who lack confidence in their mathematical ability observed in rupees were as follows:

Kanpur	15	7	11	13
Lucknow	14	10	10	6
Delhi	4	10	8	8

Do the data indicate that the price in the three cities are significantly different?

Q7.

Illustration 3: The following table gives the number of refrigerators sold by 4 salesmen in three months May, June and July:

Month	Salesman			
	A	B	C	D
March	50	40	48	39
April	46	48	50	45
May	39	44	40	39

Is there a significant difference in the sales made by the four salesmen? Is there a significant difference in the sales made during different months?

Q8. A component follows an exponential failure distribution with a failure rate of λ=0.005 failures per hour.

- (a) Find the reliability function R(t)
- (b) Determine the probability that the component will function for at least **200 hours**.
- (c) Compute the hazard function h(t.

Q9. A machine follows an exponential failure distribution with a failure rate of λ=0.001, failures per hour.

- (a) Compute the Mean Time to Failure (MTTF).
- (b) Find the probability that the machine will function beyond **1500 hours**.

Q10.

A mechanical system has a failure rate given by the Weibull hazard function:

$$h(t) = \frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1}$$

where $\beta = 1.5$ and $\eta = 5000$ hours.

- (a) Compute the hazard rate at $t = 3000$ hours.
- (b) Determine the reliability function $R(t)$ at $t = 3000$ hours.